Unit 6, Lesson 7: Applying Similar Triangles to Solve Real-World Problems

Benchmark

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Learning Target

- [ ] I can apply similar triangles to solve real-world problems.

6.7.1 Warm-Up: Crack the Code

1. Feedback is given on four combination attempts to crack the code. Use the feedback on each attempt to crack the code.

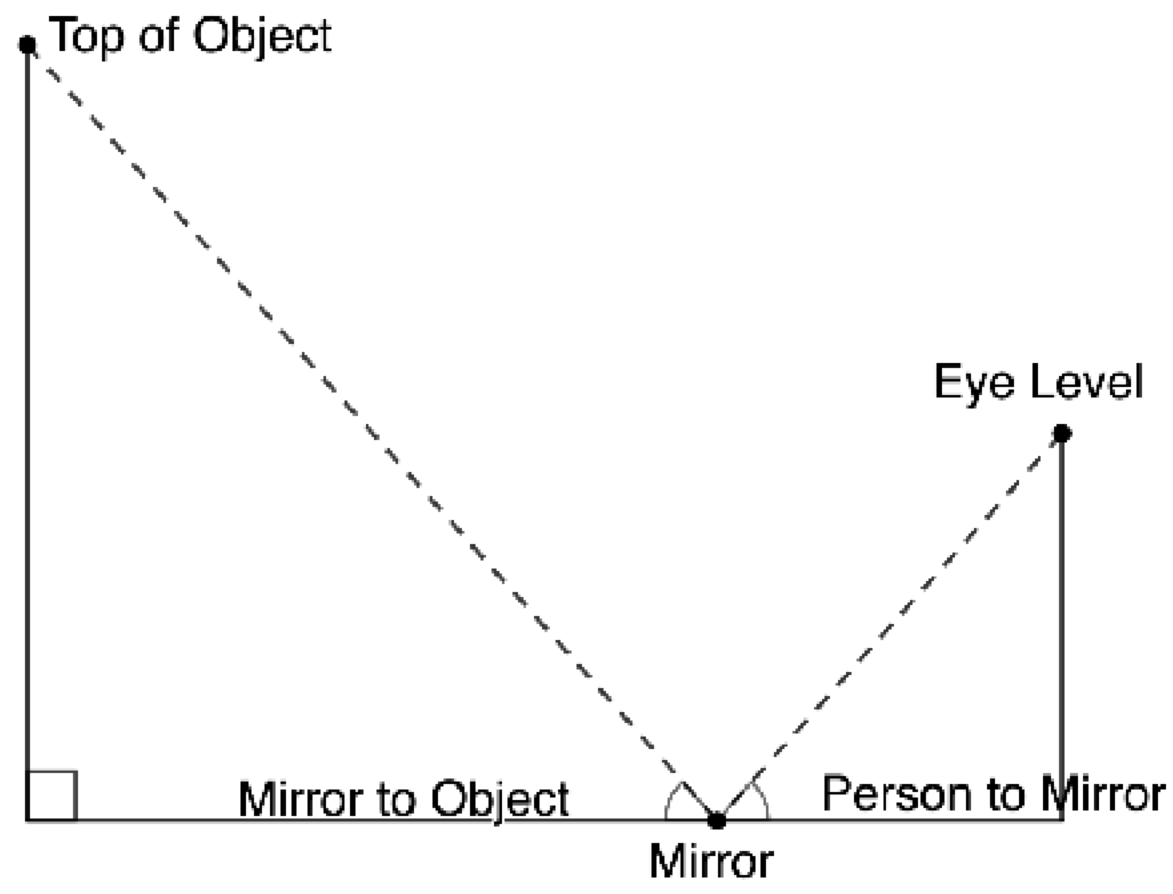
Combination Attempt			Feedback
5	6	8	One number is correct and is correctly placed.
5	8	9	Nothing is correct.
2	9	6	One number is correct, but it is in the wrong place.
4	9	1	Two numbers are correct, but they are in the wrong place.

a. What is the code?

b. Explain why your code is correct.

6.7.2 Exploration: Mirror, Mirror on the Ground

If an object is visible in a mirror positioned on the ground, then the measures of the angles of reflection from the eye to the mirror, and the mirror to the object are equal and similar triangles are created.



In this Exploration, use a mirror and similar triangles to estimate the height of objects.

Materials required: 1 yardstick per group
1 mirror per group

1. With your partner, determine who is Partner A and who is Partner B. Then, work together to follow the steps.

Step 1: Partner B measures the distance from the floor to Partner A's eyes. Label the above diagram with the measurement.

Height of eye level: _______________ inches

Step 2: Partner A places the mirror on the floor between themselves and the object to be measured, then measures the distance from the base of the object to the center of the mirror. Label the measurement on the diagram.

Step 3: Partner A starts at the mirror and walks backwards from the mirror and the object, stopping when the top of the object can be seen in the middle of the mirror.

Step 4: Partner A stays in place and Partner B measures the distance from Partner A's feet to the middle of the mirror. Label the measurement on the diagram.

Step 5: Use your knowledge of similar triangles to estimate the height of the object.

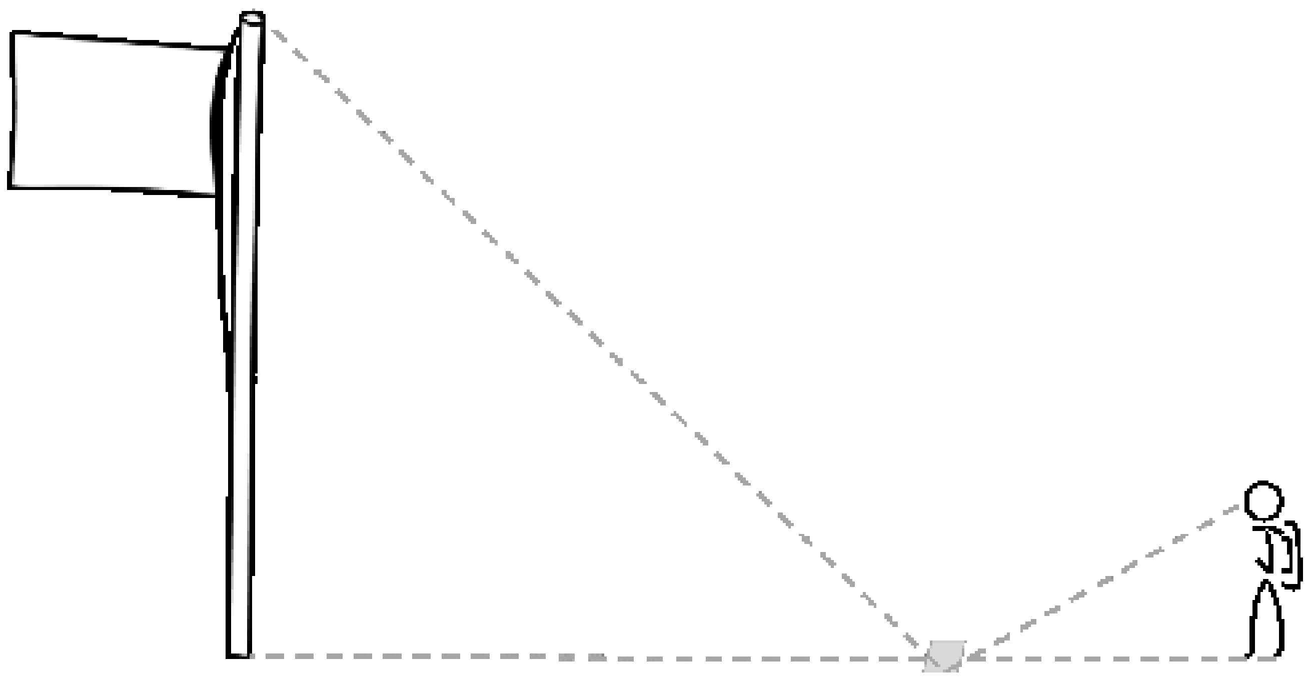
2. Repeat steps 2–5, alternating roles between Partner A and Partner B. Record your results in the table.

Object	Distance from Partner to Mirror	Distance from Object to Mirror	Work to Determine Approximate Height

6.7.3 Your Turn!

1. Mrs. Robinson assigned her class a project to find the height of a flagpole. The students could not measure the height, so they used a mirror to determine the height of the flagpole. Tayveon placed a mirror on the ground 21 feet (ft.) from the base of the flagpole and backed up until the top of the pole was centered in the mirror.

2.



Tayveon is 5.4 ft. tall and is standing 7.2 ft. from the mirror.

a. Label the diagram with the given measurements.

b. How tall is the flagpole?

3. A rock climber wants to know the height of a cliff. She places a mirror 19.55 feet (ft.) from the base of the cliff, then backs up from the mirror until she can see the top of the cliff in the center of the mirror. At that point, her eye level is 6.2 ft. above the ground and she is standing $5\frac{3}{4}$ ft. from the mirror.

a. Draw and label a diagram that represents this situation.

b. Find the height of the cliff to the nearest tenth of a foot.

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6.7.4 Wrap-Up: Reflection

1. In this unit, there were three benchmarks. For each of the benchmarks, write one thing you feel confident you know and one thing you are still unsure about.

Benchmark	One thing I am confident I know is ...	One thing I am still unsure about is ...
MA.8.AR.3.1 Determine if a linear relationship is also a proportional relationship.		
MA.8.AR.3.2 Given a table, graph or written description of a linear relationship, determine the slope.		
MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.		